

Fusion Science and Technology

ISSN: 1536-1055 (Print) 1943-7641 (Online) Journal homepage: http://www.tandfonline.com/loi/ufst20

Tritium Superpermeability: Experimental Investigation and Simulation of Tritium Recirculation in "Prometheus" Setup

R. K. Musyaev, B. S. Lebedev, S. K. Grishechkin, A. A. Yukhimchuk, A.A Busnyuk, M. E. Notkin, A. A. Samartsev & A. I. Livshits

To cite this article: R. K. Musyaev, B. S. Lebedev, S. K. Grishechkin, A. A. Yukhimchuk, A.A Busnyuk, M. E. Notkin, A. A. Samartsev & A. I. Livshits (2005) Tritium Superpermeability: Experimental Investigation and Simulation of Tritium Recirculation in "Prometheus" Setup, Fusion Science and Technology, 48:1, 35-38, DOI: 10.13182/FST05-A874

To link to this article: https://doi.org/10.13182/FST05-A874

Published online: 07 Apr 2017.

Submit your article to this journal

Article views: 5

View related articles

Citing articles: 2 View citing articles

Full Terms & Conditions of access and use can be found at
http://www.tandfonline.com/action/journalInformation?journalCode=ufst20

TRITIUM SUPERPERMEABILITY: EXPERIMENTAL INVESTIGATION AND SIMULATION OF TRITIUM RECIRCULATION IN "PROMETHEUS" SETUP

R.K. Musyaev, B.S. Lebedev, S.K. Grishechkin, A.A. Yukhimchuk, *A.A Busnyuk, *M.E. Notkin, *A.A. Samartsev, *A.I. Livshits

Russian Federal Nuclear Center-All–Russian Research Institute of Experimental Physics (RFNC-VNIIEF), 607188, Sarov, Nizhny Novgorod Region, Russi. , e-mail: arkad@triton.vniief.ru
**St.-Peterburg State University of Telecommunications (SUT), St.Peterburg, Russia*

The superpermeability phenomenon was suggested to use in fusion machines to separate the fuel mixture from helium ashes, to arrange in-vessel fuel recirculation contours and effective evacuation of fuel mixture out of machine exhaust. To develop this technology and to simulate tritium recirculation in RFNC-VNIIEF at the "Prometheus" setup the experiments on superpermeation of hydrogen isotopes through metal membrane were realized.

The results of experiments on superpermeation of hydrogen isotopes through cylindrical niobium membrane are presented. As the experiment has shown, membrane pumping rate is inversely proportional to square root of isotope mass and amount to 2.5 l/cm^{2}s for protium, 1.8 l/cm^{2}s for deuterium and 1.5 l/cm^{2}s for tritium. The possibility of effective pumping, separation of hydrogen isotopes from helium and residual gas, compression and recuperation of hydrogen isotopes by means of superpermeable membrane was demonstrated. It follows from results that the separation of D/T from He with employment of the techniques of superpermeable membranes might reduce the total amount of tritium in fuel cycle and substantially enhance the resource of cryogenic pumps evacuating helium.

I. INTRODUCTION

The term "superpermeability" taken in this paper implies that metallic membranes of a macroscopic thickness may be virtually "transparent" to suprathermal (≥ 1 eV) hydrogen particles, comparable an opening of the same area. Virtually the whole flux of gas being implanted in metal lattice, or even just sticking to metal surface, passes through the membrane, irrespective of its temperature and thickness. Hydrogen isotopes passing through the superpermeable membranes is simultaneously compressed and purged of all impurities, including helium. The superpermeability phenomenon was suggested to be made use of for an active control of fuel mixture fluxes in fusion machines, to separate the fuel from helium ashes and to arrange in-vessel fuel recirculation contours [1]. Employment of the techniques of superpermeable membranes might markedly reduce the total amount of tritium in the fuel cycle, substantially enhance the resource of cryopumps evacuating helium, and provide substantial gains with regard to both the safety and costs of the fusion devices [2].

II. EXPERIMENTAL

The "Prometheus" setup was designed to study the phenomena of the permeation and superpermeation of hydrogen isotopes, including tritium, through metals and structural materials [3, 4].

Experimental studies of the superpermeability phenomenon have been performed with the experimental cell whose scheme is shown in Fig. 1. The cell comprises

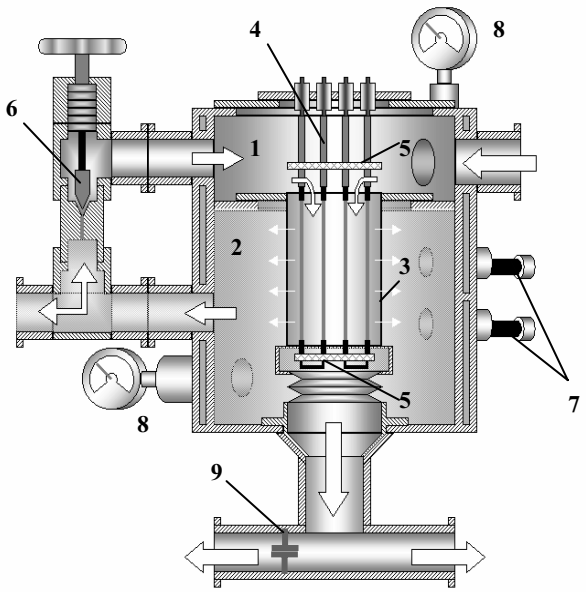


Fig. 1. The cell for research in hydrogen isotopes superpermeability: 1- input volume; 2- output volume; 3- cylindrical membrane; 4- atomizer; 5- quartz disks; 6- leak valve; 7- pyrometer; 8- pressure gauge; 9- diaphragm.

a cylindrical niobium membrane 3 under investigation (100 mm diameter, 180 mm height, and 0.1 mm wall thickness), a hydrogen atomizer 4, a recirculation connection tube, and pressure 8 and temperature gauges 7. The thermal atomizer placed inside the membrane consists of three separate tantalum ribbons that can be ohmically heated up to a temperature of 2300 K. Their total surface area amounts to 120 cm^2 (about 20% of the membrane area).

Due to the atomizer operation the membrane is heated by radiation. Its temperature is measured with pyrometers looking through special windows. The membrane is positioned vertically to divide the cell into an inlet and an outlet chambers. The inlet and outlet chambers are continually pumped with ion pumps through ducts with known conductivities. The cylindrical membrane ends screen by two quartz disks to prevent the losses of atoms produced on the atomizer inside the membrane hollow, and also to partly protect other device elements from the action of atomic hydrogen.

III. MEMBRANE PUMPING AND DEMONSTRATION OF COMPRESSION AND TRITIUM RECIRCULATION

Employment of the experimental methods developed at SPbUT is assumed in device operation [5].

The experiment is initiated with the establishment of a constant hydrogen isotope flux flowing through the membrane inlet chamber In order to suppress sorption of hydrogen isotopes and to minimize embrittlement of the atomizer ribbons, their temperature was kept in the range of 200-500°C. Heating of the atomizer ribbons to high enough temperature results in the dissociation of hydrogen molecules at the atomizer surface. The atoms generated at the atomizer absorb on the membrane inlet surface, permeate to the membrane outlet side and release into gas phase as diatomic molecules. The pressure of hydrogen isotopes in the outlet chamber is rising until a steady state pressure defined by a balance of the permeation flux and outlet chamber pumping speed is established. The pressure in the inlet chamber is decreased few order of value due to that almost the whole flow of hydrogen isotopes admitted into it passes through the membrane and into the outlet chamber. On switching off of the atomizer, the inlet and outlet pressures are returned to their starting levels. Typical experimental curves for hydrogen isotopes mixture are shown in Fig. 2.

The ability of the superpermeable membrane to compress the permeating tritium was demonstrated in following way (see Fig. 3): pumping of the outlet chamber with the external pump was ceased (point of time t_2) to result in the outlet pressure rising due to the accumulation of permeating tritium at the outlet side. That is accompanied with an inlet pressure rise because of an enhanced molecular driven permeation through the

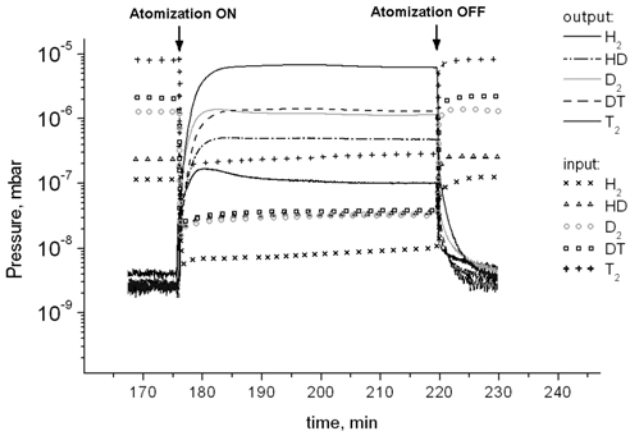


Fig. 2. A typical pattern of changes with time of the partial pressures of hydrogen isotopes at the inlet and outlet sides of the membrane that occur on switching on of the atomizer.

membrane back from the outlet to the inlet side. Comparison of the pressure increment at the input and output side was used to calculate the degree of the membrane asymmetry (relation between the inlet permeating flow and the outlet flow), which depend on degree surface passivation and in this case was ~10^3.

Since the membrane is able to compress the gas permeating through it, one also can arrange for a direct recirculation of the tritium evacuated with membrane pumping and its return into the tritium cycle. For that the outlet chamber was connected to inlet one by opening of a leak valve (6 in Fig. 1) to let the tritium compressed behind the membrane flow back into the inlet chamber (point of time t_3 in Fig.3). The leak valve was regulated so that resulted in the inlet pressure coming back to its initial value (before the atomizer initiation). Following that

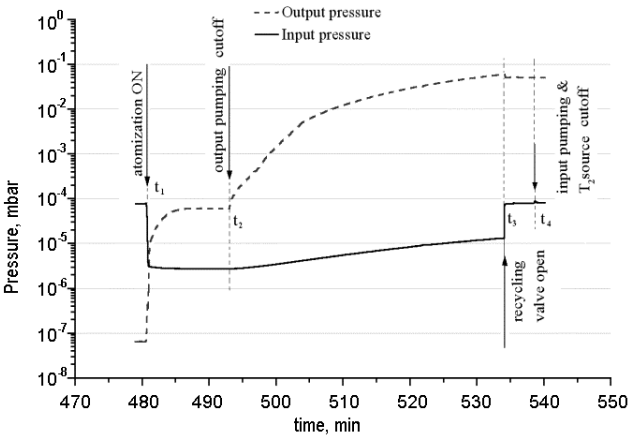


Fig. 3. Demonstration of compression and tritium recirculation by means of superpermeable membrane.

(point of time t_4), both pumping of the inlet chamber with its ion pump and the influx of tritium into it were ceased. As a result, a closed tritium circuit was formed.

The rate of hydrogen pumping by the membrane was experimentally defined by comparing it with the conductance of the duct through which the inlet chamber is pumped. As explained in our previous contributions [4], membrane pumping rate is inversely proportional to square root of isotope mass as a first approximation. Presented in Fig. 4 are experimental data on the membrane pumping of protium, deuterium and tritium as functions of inlet gas pressure (corresponding atomization rate curves for different isotopes are plotted in the same figure). One can see the curves for protium, deuterium and tritium go parallel till certain pressure. The rate of T_2 and D_2 pumping being indeed lower by a factor of about 1.7 and 1.4 respectively than the rate of the pumping of H_2 and amount to 2.5 l/cm^{2}s for protium, 1.8 l/cm^{2}s for deuterium and 1.5 l/cm^{2}s for tritium.

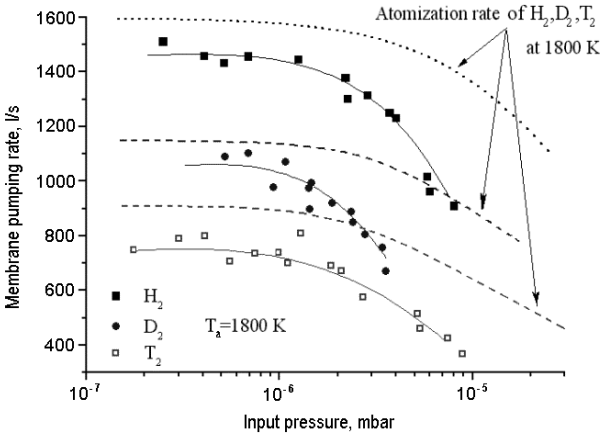


Fig. 4. Comparison of the membrane pumping of protium, deuterium and tritium

IV. TRITIUM AND HELIUM MIXTURE EXPERIMENT

The obvious separation of D/T mixtures from He is one of proposals put forward on superpermeable membranes applications in fusion [2]. A special experiment was staged to demonstrate this separation possibility of mixture. Experimental curves on time dependence of partial pressure obtained by mass-spectrometers for atomic mass 4 m/e and 6 m/e are shown in Fig.5. The experiment is initiated with the establishment of a constant gas mixture flux. After the atomizer had been switched (point of time t_1) and a permeation flux had been established, the helium flux to inlet chamber was ceased (point of time t_2). Then the atomizer was switched off (point of time t_3) and on again (point of time t_4). After second switching on, the helium flux was opened into inlet chamber again (point of time

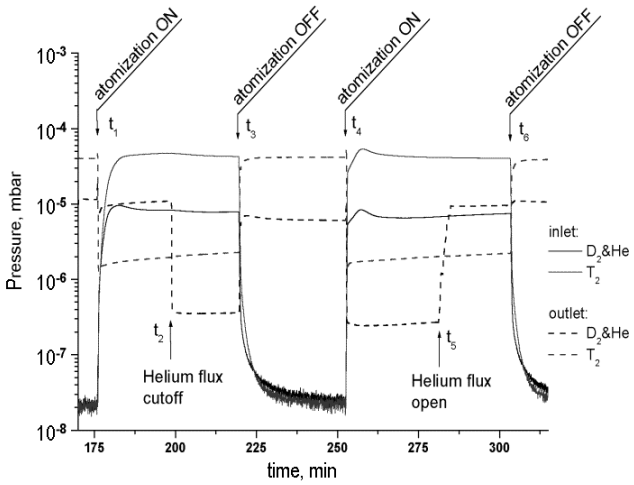


Fig. 5. Time dependence of partial pressure at the inlet and outlet sides of the membrane that occurs under change of helium flux.

t_5). As can be seen from picture the outlet pressure doesn't respond to change of inlet pressure that means of selective possibility of superpermeable membranes for hydrogen isotopes. The same pressure behaviors were occurred for radiogenic helium with mass 3 m/e.

V. TRITIUM INVENTORY

To estimate the tritium inventory in the membrane bulk the shaded area in Fig.6 was calculated for different temperatures of membrane under same initial flux. Temperature dependence of tritium retention under experimental upstream flux equals ~10^{16} D/T per second during continual evacuation of outlet chamber is presented in Fig.7. This retention means a quantity of

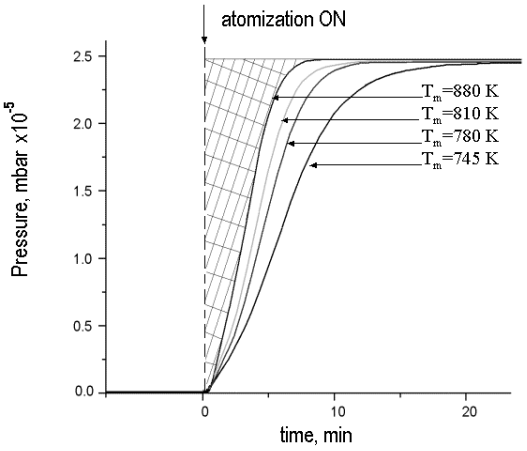


Fig. 6. Time dependence of tritium pressure at the outlet side of the membrane for different membrane temperatures.

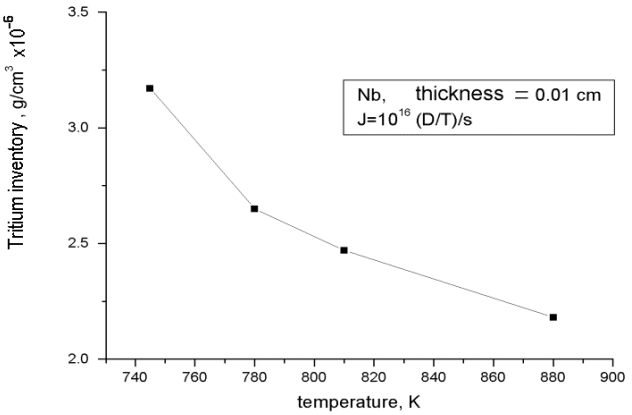


Fig. 7. Temperature dependence of tritium inventory to cm^3 of membrane.

tritium per cubic cm of metal membrane. This result is in accord with calculated evaluation value from work [6] that point to a possible way to decrease the tritium inventory in the membrane by means of operation at higher temperature, where superpermeable regime still exists [7].

VI. CONCLUSION

The superpermeability of tritium through niobium membrane was experimentally observed. As experience has shown the membrane pumping rate is inversely as the square root of the isotopes mass.

The possibility of effective pumping, separation from helium and residual gas, compression and recirculation of hydrogen isotopes particular D/T mixture by means of superpermeable membrane as well as tritium inventory decrease under temperature increasing was demonstrated. Design and application of superpermeable membrane pumps at closed cycle permit to economically use heavy hydrogen isotopes in research activities.

REFERENCES

[1] A.I. LIVSHITS, M.E. NOTKIN, A.A. SAMARTSEV, "Physico-chemical origin of superpermeability – large-scale effects of surface chemistry on "hot" hydrogen permeation and absorption in metals," *J.Nucl.Mater*. **170**, 79–94 (1990).

[2] A. I. LIVSHITS, M.E. NOTKIN, V.I. PISTUNOVICH et al., "Superpermeability: Critical points for Applications in Fusion", *J.Nucl.Mater*. **220-222**, 259 (1995).

[3] S.K. GRISHECHKIN, A.N. GOLUBKOV, E.V. GORNOSTAEV et al., "Device for investigations of tritium retention in and permeation through metals and structural materials,". *Hydrogen Recycling at Plasma Facing Materials, NATO Science Series, II Mathematics, Physics and Chemistry*, Vol.**1**, 339-348 (2000).

[4] A.A.YUKHIMCHUK, R.K. MUSYAEV, YU.I. VINOGRADOV et al., "Prometheus" setup for study of tritium superpermeation", *J.Fusion Science and Technology*, Vol.**41**, 929-933 (2002).

[5] A.I. LIVSHITS, M.E. NOTKIN, A.A. SAMARTSEV et al., "Superpermeability to fast and thermal hydrogen particles: Applications to the pumping and recycling of hydrogen isotopes", *J.Nucl.Mater.* **196–198**, 159 (1992).

[6] A.I. LIVSHITS, Y. HATANO and K WATANABE, "Superpermeability in the fusion technology: tritium accumulation and compression", *J.Fusion Science and Technology*, Vol.**41**, 882-886 (2002).

[7] A.I. LIVSHITS, "Superpermeability of solid membranes and gas evacuation. Part I. Theory", *Vacuum*, **29**, 103 (1979).